

Yukith M Joseph

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EDUCATION

Presidency University, Bengaluru - B.Tech, Computer Science & Engineering (Networks) 2024 - 2028 (Exp.)
CGPA: 7.56/10 | Coursework: Computer Networks, Network Security, AI, DBMS, Data Structures & Algorithms

EXPERIENCE

- Full-Stack Development Intern** | Elevance Skills | MERN, Next.js, TypeScript, Razorpay Jun 2026 (1 month)
• Built YukithHub, a full-stack community platform (Next.js, React, TypeScript, Node.js/Express, MongoDB) with JWT/bcrypt auth, email and SMS OTP verification, and password recovery; GitHub.
- Implemented device-based login rules with login history, Razorpay subscription tiers with emailed invoices, a reward-points system, and six-language support with OTP-gated switching.
- Capstone Project** | Presidency University x NVIDIA | Python, PyTorch, Hugging Face, Streamlit Jun 1 - Jun 10, 2026
• Built AUTOHIRE, a local-first AI hiring platform: PyMuPDF resume parsing, Sentence Transformers (all-MiniLM-L6-v2, CUDA with CPU fallback) for semantic job matching plus deterministic ATS scoring, with no external APIs; GitHub.
- Delivered a Candidate Portal (job recommendations, multi-step assessments) and a Recruiter Console (applicant pipeline, AI reports, offer-letter generation).

PROJECTS

- Q-SHIELD** | Python, FastAPI, OpenCV, YOLO11n, ML-KEM/ML-DSA; hackathon prototype GitHub
• Built a continuous device-trust gateway that scores each IoT device from six weighted evidence factors (HMAC-SHA256 identity, tamper switch, sensors, config integrity, liveness, camera) and quarantines it when trust drops.
- Implemented post-quantum cryptography: camera observations are signed over an ML-KEM-768 (FIPS 203) session and the SHA-256-linked evidence chain is signed with ML-DSA-65 (FIPS 204); verified against NIST ACVP test vectors.
- Built an OpenCV + YOLO11n vision service with camera-tamper checks (covered, frozen, redirected view), a digital twin of expected vs. reported device state, and an operator-verified recovery flow.
- JIVA** | TypeScript, Express, Socket.IO, Zod, MapLibre, AWS CDK GitHub
• Designed an event-driven healthcare coordination platform (monorepo, 9 shared packages, 4 role-based web apps) linking ambulances, hospitals and patients over REST and Socket.IO without exposing hospital databases.
- Implemented a Hospital Acceptance Protocol with automatic re-routing on rejection/timeout, provider-agnostic routing (Valhalla, OSRM, fallback), and confidence/provenance on every derived value; 20 adversarial failure scenarios pass; AWS architecture modeled in CDK (not deployed).
- ETTH - Encrypted Traffic Threat Hunter** | Python, scikit-learn, dpkt/Scapy GitHub
• Built an ML pipeline that detects threats in encrypted traffic without decrypting payloads: bidirectional flow reconstruction from raw PCAPs and TLS ClientHello fingerprinting (JA3, JA3S, JA4).
- Prevented data leakage with GroupShuffleSplit train/test isolation and removal of deterministic identifiers; compared five experimental configurations, documented dataset confounding, and covered the code with 46 unit tests.
- ALIAS - Login Anomaly Investigation** | Python, FastAPI, SQLAlchemy, React, WebSocket, Gemini GitHub
• Built a login-security platform with six behavioral baseline models (temporal, device, location, network, auth, access), multi-signal correlation and risk scoring; Gemini generates investigation reports; real-time WebSocket alerts to a React SOC dashboard; 61 backend tests pass.
- Sugar AI - Offline Voice Assistant** | Python, faster-whisper, Ollama, MeloTTS, WebRTC VAD GitHub
• Built a fully local voice assistant: faster-whisper (large-v3-turbo) speech-to-text, Ollama-hosted LLMs, MeloTTS output, and VAD-based wake/sleep handling in a multithreaded pipeline, with Spotify API and desktop automation integration.
- T R Constructions Website** | Next.js 16, TypeScript, Tailwind CSS v4, GSAP, React Three Fiber Live site
• Built and deployed a production site for a Bengaluru civil contractor on Vercel, with a typed content layer, static CSP/security headers, and a rate-limited, honeypot-protected enquiry endpoint.

ACHIEVEMENTS

- **Smart India Hackathon 2025** - Top 30 Finalist, national level (Government of India)
- **Nirmith 2026** - National Finalist, NITTE Meenakshi Institute of Technology
- **InnovateX 4.0 Hardware Hackathon and Hardware Expo** - Runner-Up (both), Presidency University
- **Prompt Wars**, Presidency University - Top 15 of 180 teams
- Participant: ISAKA Hackathon; Odoo x NMIT Hackathon 2026 | **Certification:** Microsoft Azure AI Fundamentals (AI-900)

LEADERSHIP

Student Coordinator, 30-hour AWS Innovation Challenge - Presidency University

Organiser, InnovateX Tech Fest: 10 concurrent events, ~280 participants, 60-70 teams; World Record 2026

Event Head, Build Club | **Core Coordinator**, One-O-One Club | **Event Head**, Informatica Club 2024 - Present

TECHNICAL SKILLS

Languages: Python, C, Embedded C, TypeScript/JavaScript, SQL

AI/ML: PyTorch, scikit-learn, YOLO (Ultralytics), OpenCV, faster-whisper, Ollama, Gemini API, XGBoost

Backend/Cloud: FastAPI, Express, MongoDB, Next.js, React, WebSockets/Socket.IO, SQLite, JWT, Vercel, AWS CDK

Security/Networking: TCP/IP, TLS, JA3/JA4, PCAP analysis (Scapy/dpkt), post-quantum crypto (ML-KEM, ML-DSA), access control

Embedded/Tools: Arduino, ESP32, Git, pytest, Unity